

#1.List Tasks

#1 Create a list of 5 student names.

```
students = []

for i in range(5):
    name = input("Enter student name: ")
    students.append(name)

print("Student List:", students)
```

#2 Add a new name to the list.

```
s=['a','b','c']
s.append('d')
print(s)
```

#3 Remove a student name from the list.

```
s=['a','b','c']
s.remove('a')
print(s)
```

#4 Print all elements using loop.

```
students = ["Arun", "Bala", "Charan", "Deepak", "Esha"]
```

```
for name in students:
```

```
    print(name)
```

#5 Find largest and smallest number in list.

```
s=[1,324,32,23,12]
```

```
print("Largest no:", max(s))
```

```
print("smallest no:", min(s))
```

#6 Count even and odd numbers in list.

```
s=[1,324,32,23,12]
```

```
odd=0
```

```
even=0
```

```
for i in s:
```

```
    if i%2==0:
```

```
        even+=1
```

```
    else:
```

```
        odd+=1
```

```
print("no. of even",even)
```

```
print("no. of even",odd)
```

#7 Reverse a list.

```
s=[1,2,3,4]
```

```
s.reverse()
```

```
print(s)
```

#8 Sort a list in ascending order.

```
s=[1,2,3,4]
```

```
s.sort()
```

```
print(s)
```

#9 Find sum and average of list elements.

```
s=[1,2,3,4]
```

```
total=sum(s)
```

```
average=total/len(s)
```

```
print("total:",total)
```

```
print("average:",average)
```

#10 Search an element in list.

```
s=[1,2,3,4]
```

```
x=int(input("enter a number to found:-"))
```

```
if x in s:
```

```
    print("element found")
```

else:

```
print("element not found")"
```

#2.tuple

#1 Create a tuple with 5 numbers.

```
t=(1,2,3,4)
```

#2 Access tuple elements using indexing.

```
"print(t[0])
```

```
print(t[3])"
```

#3 Find length of tuple.

```
"print(len(t))"
```

#4 Count occurrence of an element.

```
"a=int(input("enter a number in tuple"))
```

```
print("Occurrence of", a, ":", t.count(a))"
```

#5 Find index position of element.

```
""a=int(input("enter a number in tuple"))  
print(t.index(a))""
```

#6 Slice a tuple.

```
""print(t[1:4])""
```

#7 Convert tuple into list

```
""b=list(t)  
print(b)""
```

#8 Convert list into tuple.

```
""l=[1,2,3,4]  
b=tuple(l)  
print(b)""
```

#9 Find maximum and minimum value.

```
""print(max(t))  
print(min(t))""
```

#10 Iterate tuple using loop.

```
'''print("Tuple elements:")
```

```
for i in t:
```

```
    print(i)'''
```

```
#Set Tasks
```

```
#1 Create a set of numbers.
```

```
s={10,20,30,40}
```

```
#2 Add elements into set.
```

```
'''a=int(input("enter a number to add"))
```

```
s.add(a)
```

```
print(s)'''
```

```
#3 Remove elements from set.
```

```
'''a=int(input("enter a number to remove"))
```

```
s.remove(a)
```

```
print(s)'''
```

```
#4 Find union of two sets.
```

```
t={50,60,70,80}
```

```
""print(s|t)""
```

#5 Find intersection of two sets.

```
""print(s&t)""
```

#6 Find difference between sets.

```
""print(s-t)""
```

#7 Remove duplicates from list using set.

```
""lst = [1, 2, 2, 3, 4, 4, 5]
```

```
unique_list = list(set(lst))
```

```
print("List without duplicates:", unique_list)""
```

#8 Check subset and superset.

```
""x = {1, 2}
```

```
y = {1, 2, 3, 4}
```

```
print("x is subset of y:", x.issubset(y))
```

```
print("y is superset of x:", y.issuperset(x))""
```

#9 Find unique vowels in string.

```
""s1 = str(input("enter a word"))
```

```
vowels = {'a', 'e', 'i', 'o', 'u'}  
  
unique_vowels = set(s1) & vowels  
  
print("Unique vowels:", unique_vowels)"
```

#10 Convert list into set.

```
"d=["s","d","c"]  
  
e=set(d)  
  
print(e)"
```

#dictionary

#1 Create a student dictionary with name and marks.

```
"student = {  
  
    "name": "Rahul",  
  
    "marks": 85  
  
}  
  
print("Student Dictionary:", student)"
```

#2 Add new key-value pair.

```
"student["grade"] = "A"
```



```
print("After adding new key-value:", student)"""
```

#3 Update dictionary value.

```
"""student["marks"] = 90
```

```
print("After updating marks:", student)"""
```

#4 Delete an item from dictionary.

```
"""del student["grade"]
```

```
print("After deleting an item:", student)"""
```

#5 Print all keys and values.

```
"""print("Keys:")
```

```
for key in student.keys():
```

```
    print(key)"""
```

```
print("Values:")
```

```
for value in student.values():
```

```
    print(value)
```

